Khanh Nguyen, Postdoc, Ph.D.

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EDUCATION

Doctor of Philosophy in Smart Vehicle Engineering
Konkuk University (KU), Seoul, Republic of Korea (Feb. 2022 – Mar. 2026)
GPA: 4.0/4.0
Dissertation: Scaling-Based Design Approach for Tailbeat Fish-Inspired Robots Swimming at High Speed and Propulsive Efficiency

Master of Science in Smart Vehicle Engineering
Konkuk University (KU), Seoul, Republic of Korea (Mar. 2019 – Feb. 2021)
GPA: 3.9/4.0
Thesis: Investigation of stability and aerodynamic performance of a flapping-wing micro air vehicle in hover using 3D computational fluid dynamics (CFD) analyses

Bachelor of Engineering in Mechanical and Aerospace Engineering
Vietnam National University, Ho Chi Minh University of Technology (HCMUT), Ho Chi Minh, Vietnam (Sep. 2013 – Jun. 2018)
Frech – Vietnamese Training Program of Excellent Engineers (PFIEV)
The highly selective program is accredited by France's Engineering Degree Commission (2004 – 2022) and designated as a EUR – ACE Master Program by the European Network for Accreditation of Engineering Education (2010 – 2022), comprising 274 Credits.
GPA: 3.2/4.0
Thesis: Computational Approach on the Aerodynamics of UAV combining fixed wing and three propellers (Grade: 9.1/10)

RESEARCH

Flapping-Wing Micro Air Vehicle in Fast Forward Flight (2024)

- Processed the measured wing kinematics under tethered condition with a tilting stroke plane at an inflow speed of 4.3 m/s.

Development of Biomimetic Underwater Robot Platform (2022)

- Developed a robotic fish that achieved a speed of 1.79 m/s with a low cost of transport using a scaling-based design approach.
- Developed a larger-scale model for faster speed and higher-payload capability.
- Conducted simulation to study hydrodynamic force production and vortex formation using measured tailbeat swimming kinematics.

Toward Flapping Flights on Mars (2021)

- Investigated dynamic stability characteristics of a flapping-wing hovering on Mars.
- Analyzed aerodynamics of a flapper during takeoff under ultra-low air densities.

Leaping Robotic Fish (2020)

- Analyzed the feasibility of gliding in a flying-fish-like robot after water exit.
- Estimated undulatory body drag of robotic fish.

Aerodynamic improvement of a hovering FW-MAV (2019)

- Investigated the aerodynamic effects of spanwise corrugation, wing rotation angles, and clap-fling mechanism.
- Proposed optimal wing kinematics that improved aerodynamic efficiency by 31%.

Comparative stability analyses of FW-MAVs (2019)

- Compared the longitudinal and lateral stability characteristics of two flapping-wing robots.

Aerodynamic Characteristics of a Propeller-Driven Fixed-Wing Aircraft During Takeoff and Forward Flight (2018)

- Analyzed laminar purple separation along chordwise and spanwise positions of fix wings for different wing angle of attack.
- Estimated the tip-path-plane angle of the tricopter in forward flight using CFD-derived aerodynamic force coefficients.
- Analyzed aerodynamics of a tricopter frame during takeoff using the Virtual Blade Method in OpenFOAM.

SERVICES & MENTORSHIP

Professional Services

- **Journal referee,** Journal of Fluids and Structures, Chinese Journal of Aeronautics, Bioinspiration & Biomimetics, Engineering Research Express, Journal of Aeronautics Astronautics and Aviation, International Journal of Intelligent Unmanned Systems.

Academic Services & Teaching Assistant

- **Grading Assistant,** Basics of Mechanics (KU); Finite Element Method (KU); Fluid Mechanics (HCMUT).
- **Thesis Evaluator,** Graduation theses of Bachelor's students (HCMUT, 2026)

Research & Mentorship Assistant

- **Co-supervisor,** Undergraduate research and capstone projects, Dr. Thi Hong Hieu Le's Lab, HCMUT (2025-present).
- **Mentor,** Master's students in CFD simulations and mechanical design, Dr. Taesam Kang's Lab, KU (2024).
- **Advisor,** Undergraduate capstone projects, Dr. Thi Hong Hieu Le's Lab, HCMUT (2018).

AWARDS AND FELLOWSHIP

- Excellent Dissertation Award, Korean Society of Mechanical Engineers (Only International Recipient in Top 5 Nationwide; 2026)
- Brain Korea 21 Scholarship, Korean Government Program Supporting Outstanding Scholars ($23,000 for postdoc; 2026)
- Postdoctoral Fellowship, Konkuk University ($33,000; 2026)
- Academic Award for Outstanding Ph.D. Research Achievement (1st Place), Konkuk University (2026)
- Best Paper Award at Korea Society for Aeronautical and Space Sciences Conference (2024)
- Best Paper Award at International Conference on Intelligent Unmanned Systems (2022 & 2025)
- Ph.D. Fellowship, Konkuk University ($78,000; 2022 – 2026)
- Research Assistant Fellowship, Ho Chi Minh University of Technology ($1,200; 2018) and KU ($22,000; 2019 – 2021)
- Merit-based Scholarship, Konkuk University (50% Tuition; $21,000 over 7 semesters; 2019 – 2024)
- Teaching Assistant Fellowship, Ho Chi Minh University of Technology ($100, 2018) and KU ($10,000; 2024)
- Excellent Student of Ho Chi Minh University of Technology (150% Tuition; 2018)
- Quintessential Student of Ho Chi Minh University of Technology (125% Tuition; 2014 & 2017)
- Quintessential Student of Tran Phu High School in National University Entrance Exam (Score: 25/30, Top 1% Nationwide; 2013)

TECHNICAL SKILLS

- Programming Languages: MATLAB, C.
- Development Tools: Visual Studio, VS Code.
- Simulation and Post-Processing Tools: ANSYS-Fluent, CFD-Post, OpenFOAM, ParaFoam.
- Tools: CNC & 3D Printing Machines, Direct Linear Transformation Digitizing Tool.
- Software: AutoCAD, Adobe Photoshop, MS Office, SolidWorks.
- Processes: Silicone Mold Making, Transducer Measurement, Image Processing.

PUBLICATIONS

Journal Papers

1. Khanh Nguyen, Hoon Cheol Park. Scaling-Based Design Approach for Tailbeat Fish-Inspired Robots Swimming at High Speed and Propulsive Efficiency. Submitted to Ocean Engineering, September 2025; under third-round peer review.
2. Giheon Ha, Khanh Nguyen, Yu Jinyu, Teasam Kang, Hoon Cheol Park. Can flapping wings keep lift coefficient unchanged at takeoff under extremely low air density? Submitted to Acta Astronautica, March 2026; under second-round peer review.
3. Thi Hong Hieu Le, Khanh Nguyen, Vuong Thi Hong Nhi. Numerical analysis for aerodynamic characteristics of the unmanned aerial vehicle (UAV) in forward flight. J. Aeronaut. Astronaut. Aviat. 1081 2024. (JCR Q3, IF = 1.1)
4. Khanh Nguyen, Giheon Ha, Teasam Kang, Hoon Cheol Park. Analysis of hovering flight stability of an insect-like flapping-wing robot in Martian condition. Aerosp. Sci. Technol. 152 109371 2024. (JCR Q1, IF = 5.8, Rank: 6/55 in Aerospace Engineering; Link)
5. Khanh Nguyen, Hoon Cheol Park, Feasibility study on mimicking the tail-beating supported gliding flight of flying fish. Ocean Eng. 287 115745 2023. (JCR Q1, IF = 4.6, Rank: 2/25, Top 8% in Marine Engineering; Link)
6. Tan Hanh Pham, Khanh Nguyen, Hoon Cheol Park. A robotic fish capable of fast underwater swimming and water leaping with high Froude number. Ocean Eng. 268 113512 2023. (JCR Q1, IF = 4.6, Rank: 2/25, Top 8% in Marine Engineering; Link)
7. Khanh Nguyen, Loan Thi Kim Au, Hoang Vu Phan, Hoon Cheol Park. Comparative dynamic flight stability of insect-inspired flapping-wing micro air vehicles in hover: Longitudinal and lateral motions. Aerosp. Sci. Technol. 119 107085 2021. (JCR Q1, IF = 5.5, Rank: 2/34, Top 6% in Aerospace Engineering; Link)
8. Khanh Nguyen, Loan Thi Kim Au, Hoang-Vu Phan, Soo Hyung Park, Hoon Cheol Park. Effects of wing kinematics, corrugation, and clap-and-fling on aerodynamic efficiency of a hovering insect-inspired flapping-wing micro air vehicle. Aerosp. Sci. Technol. 2021. (JCR Q1, IF = 5.5, Rank: 2/34,Top 6% in Aerospace Engineering; Link)
9. Doan Kim Khanh Tran, Khanh Nguyen, Thi Hong Hieu Le, Ngoc Hien Nguyen. Numerical simulation for the forward flight of the tri-copter using virtual blade model. J. Adv. Res. Fluid Mech. Therm. Sci. 67 1 1-32 2020. (SJR Q3, IF = 0.3)

Conference Papers

1. Khanh Nguyen, Giheon Ha, Hoon Cheol Park, Design and fabrication of high-thrust tail-beating mechanism for fish-inspired swimming robot, ICIUS, Indonesia, 2025. (Presenter & Best paper award).
2. Giheon Ha, Khanh Nguyen, Hoon Cheol Park, A study on the takeoff of an insect-like flapping-wing system under low air density and low gravity conditions, Proceedings of KSAS, Korea, 2024. (Best paper award)
3. Khanh Nguyen, Giheon Ha, Hoon Cheol Park, Design and fabrication of high-thrust tail-beating mechanism for fish-inspired swimming robot, ICIUS, Indonesia, 2024. (Presenter)
4. Khanh Nguyen, Hoon Cheol Park, Analytical and experimental performance verifications of a fast-swimming robotic fish, ICIUS, Indonesia, 2024. (Presenter)
5. Khanh Nguyen, Kang Teasam, Hoon Cheol Park, Hovering characteristics of an insect-like flapping-wing robot on Mars, Proceedings of KSAS, Korea, 2023. (Presenter)
6. Khanh Nguyen, Giheon Ha, Hoon Cheol Park, Preliminary design of a fish-like fast robot by scaling of the KUFish, ICIUS, AU, 2023.
7. Khanh Nguyen, Hoon Cheol Park, Roles of hydrodynamic forces generated by tail-beating motion in gliding flight of flying-fish-mimicking robot, ICIUS, Adelaide, Australia, 2023. (Presenter)
8. Khanh Nguyen, Tan Hanh Pham, Hoon Cheol Park, Numerical investigation of hydrodynamics for a fish-like robot under undulatory forward swimming, Proceedings of the Korean Society of Mechanical Engineers Annual Meeting, Jeju, Korea, 2022. (Presenter)
9. Tan Hanh Pham, Khanh Nguyen, Hoon Cheol Park, Leaping out of water of the KUFish: Prediction and demonstration, ICIUS, Japan, 2022. (Best paper award).
10. Khanh Nguyen, Tan Hanh Pham, Hoon Cheol Park, Numerical estimation of hydrodynamic thrust using the measured tail-beating kinematics of a fish-like robot, ICIUS, Japan, 2022. (Presenter)
11. Khanh Nguyen, Loan Thi Kim Au, Hoang Vu Phan, Hoon Cheol Park, Wing kinematics modulation in an insect-like tailless flapping wing micro air vehicle (FW-MAV) for higher aerodynamic efficiency, ICIUS, Vietnam, 2021. (Presenter)
12. Khanh Nguyen, Loan Thi Kim Au, Hoon Cheol Park, Three-dimensional wing kinematics for improved aerodynamic performance of insect-like flapping-wing micro air vehicle, Proceedings of KSAS, Korea, 2020. (Presenter)
13. Loan Thi Kim Au, Khanh Nguyen, Soo Huyn Park, Hoon Cheol Park, Effect of wing corrugation on aerodynamic performance in 3D flapping wings, Proceedings of KSAS, Korea, 2019. (Presenter)
14. Doan Kim Khanh Tran, Khanh Nguyen, Thi Hong Hieu Le, Numerical simulation for the forward flight of the tri-copter using Virtual Blade Model, SAWAE, Malaysia, 2019.
15. Khanh Nguyen, Ngoc Hien Nguyen, Thi Hong Hieu Le, Numerical approach for the vertical take-off and landing UAVs using the virtual blade model, SAWAE, Thailand, 2018. (Presenter)